Lesson 4.3 - The Coefficient of Quartile Deviation

> **Video:** How To Calculate the Coefficient of Quartile Deviation by The Organic Chemistry Tutor **Chapter:** 4 - Measures of Position

Formulas

Location of the k-th quartile in the sorted data:

$$\text{position of } Q_k = \frac{k(n+1)}{4}$$

The formula gives the **position**. Read the **value** from the data.

Quantity	Formula
Interquartile range	$\text{IQR} = Q_3 - Q_1$
Quartile deviation (semi-interquartile range)	$\text{QD} = \frac{Q_3 - Q_1}{2}$
Coefficient of quartile deviation	$\text{CQD} = \frac{Q_3 - Q_1}{Q_3 + Q_1}$

Example

The data set has $n = 11$ values (already sorted). Its 3rd value is 14 and its 9th value is 35.

Find the quartiles.

- Q_1: position $\frac{k(n+1)}{4} = \frac{1(11+1)}{4} = 3$ → 3rd value → $Q_1 = 14$
- Q_3: position $\frac{k(n+1)}{4} = \frac{3(11+1)}{4} = 9$ → 9th value → $Q_3 = 35$

(a) Interquartile range

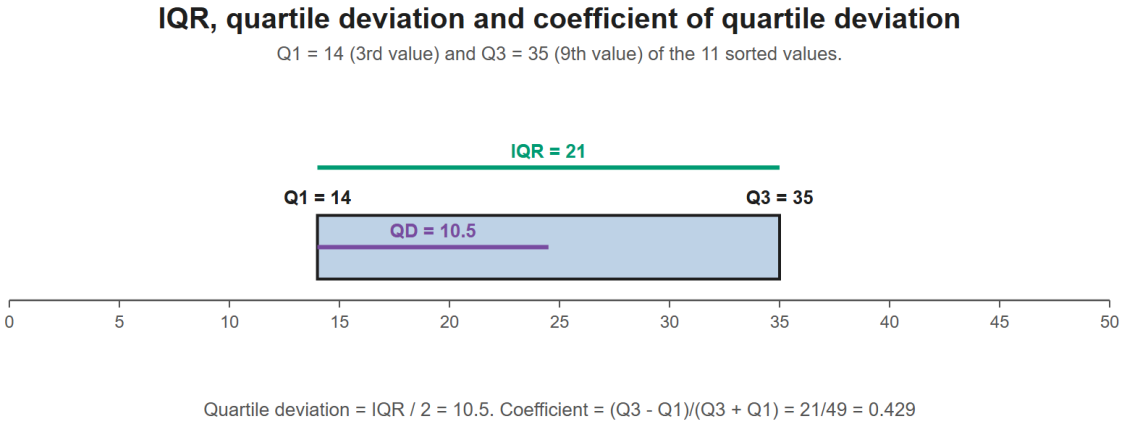
$$\text{IQR} = Q_3 - Q_1 = 35 - 14 = 21$$

(b) Quartile deviation

$$\text{QD} = \frac{Q_3 - Q_1}{2} = \frac{21}{2} = 10.5$$

(c) Coefficient of quartile deviation

$$\text{CQD} = \frac{Q_3 - Q_1}{Q_3 + Q_1} = \frac{35 - 14}{35 + 14} = \frac{21}{49} \approx 0.429$$



The IQR is the width of the middle half of the data (35 - 14 = 21). The quartile deviation is half of that (10.5). The coefficient of quartile deviation divides the IQR by Q3 + Q1, which removes the units.

Key takeaways

- QD = half the IQR. It is a spread measure that ignores the extreme 25% at each end.
- CQD = $(Q_3 - Q_1)/(Q_3 + Q_1)$ has **no units**, so it can compare spread across data sets, like the coefficient of range ($(H - L)/(H + L)$, Lesson 3.4).